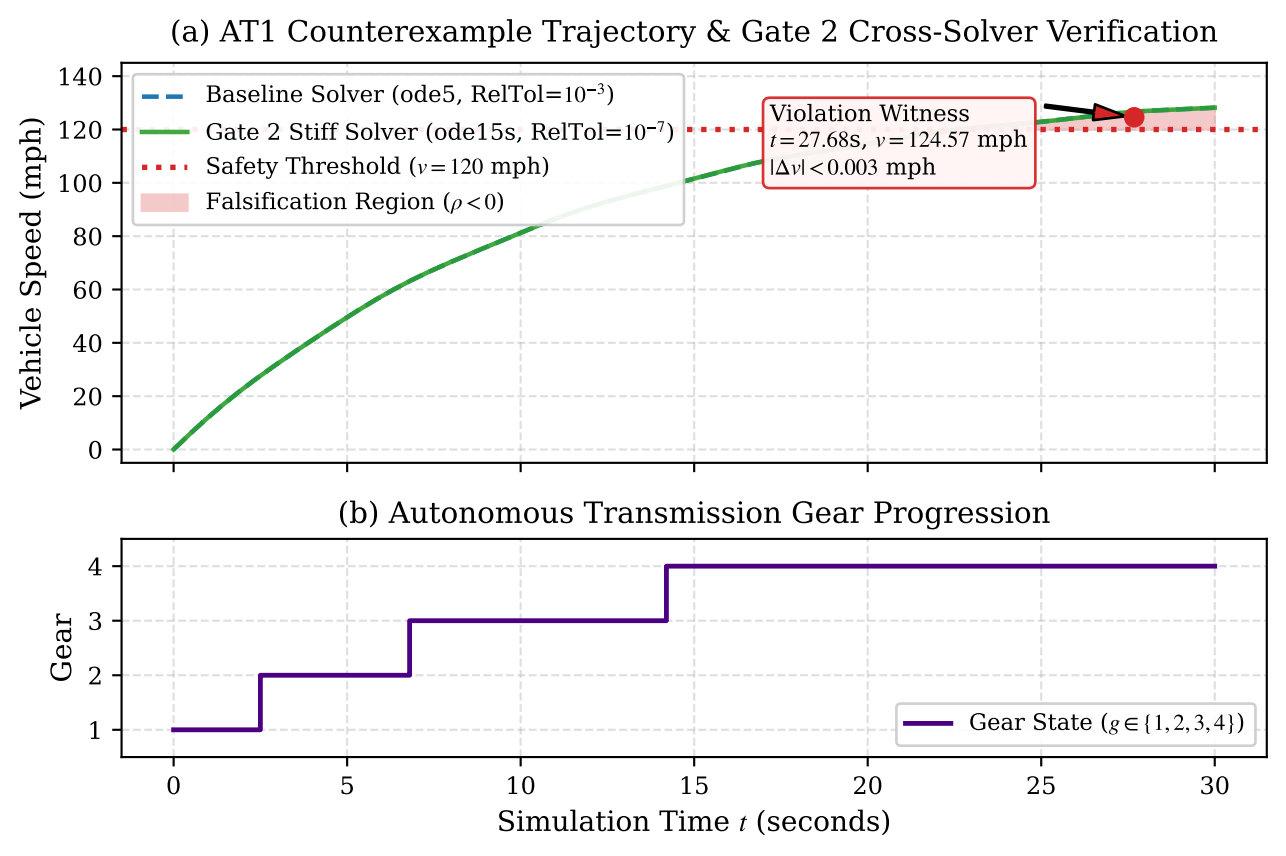
(a) AT1 Counterexample Trajectory & Gate 2 Cross-Solver Verification
Vehicle Speed (mph)
Baseline Solver (ode5, RelTol=10⁻³)
Gate 2 Stiff Solver (ode15s, RelTol=10⁻⁷)
Safety Threshold (v = 120 mph)
Falsification Region (ρ < 0)
Violation Witness
t = 27.68s, v = 124.57 mph
|Δv| < 0.003 mph
(b) Autonomous Transmission Gear Progression
Gear
Gear State (g ∈ {1, 2, 3, 4})
Simulation Time t (seconds)